

Emission Governor Handshake & Inspection Card

(Registers, handshake, invariants, and REG_ALL debug block)

Registers & knobs. Spatial registers $R = (R_x, R_y, R_z)$; budgets $B = (B_x, B_y, B_z)$; load counter $L \geq 0$; capacity $K_{\text{cap}} > 0$; normalized fill $\rho_{\text{emit}} = \frac{L}{K_{\text{cap}}}$; coherence $C \in [0, 1]$; entropy $S \geq 0$. Symmetry gate $\sigma \in \{0, 1\}^3$ (legal turns). Damping $\gamma \in (0, 1)$ (“sea”). Emit drains: $\lambda\varepsilon$; boosts: $\Delta C, \Delta S \geq 0$.

Costed kick (per tick). Apply symmetry then spend budget:

$$\Delta R \mapsto \sigma \odot \Delta R, \quad \Delta L = |B_x \Delta R_x| + |B_y \Delta R_y| + |B_z \Delta R_z|, \quad L \leftarrow (1 - \gamma)L + \Delta L.$$

Emission rule (deterministic).

$$\rho_{\text{emit}} \geq 1 \implies \text{emit one cooling quantum of energy } \varepsilon,$$

$$L \leftarrow \max(0, L - \lambda\varepsilon), \quad C \leftarrow \min(1, C + \Delta C), \quad S \leftarrow \max(0, S - \Delta S),$$

iterate while $\rho_{\text{emit}} \geq 1$ (overshoot-safe).

Soft-edge (optional).

$$p_{\text{emit}} = \frac{1}{1 + e^{-k(\rho_{\text{emit}} - 1)}} \quad (\text{emit Bernoulli with prob. } p_{\text{emit}}; \text{ at most once per tick}).$$

State sketch. IDLE $\rightarrow$ ACCUMULATE $\xrightarrow{\rho < 1}$ ACCUMULATE $\mid \xrightarrow{\rho \geq 1}$ EMIT $\rightarrow$ COOLDOWN $\rightarrow$ ACCUMULATE/IDLE.

Invariants (quick checks). (i) *No-churn*: with $\Delta R = 0$, $L \downarrow 0$ in finite time and emits stop. (ii) *Budget accounting*: $\text{work}_{\text{in}} \approx L + \lambda \sum \varepsilon_{\text{emitted}}$ (up to damping losses). (iii) *Symmetry*: if $\sigma_i = 0$ then axis i contributes zero to ΔL .

Symmetry probes (axis/diagonal). Axes: $(\pm 1, 0, 0)$, etc.; face diagonals: $(\pm 1, \pm 1, 0)/\sqrt{2}$; body diagonals: $(\pm 1, \pm 1, \pm 1)/\sqrt{3}$. Parity: $Q(u) \approx Q(-u)$. Cubic classing: within each class, all permutations match within tolerance. Corner test (equal engagement): along $(1, 1, 1)/\sqrt{3}$, registers contribute equally.

REG_ALL (atomic inspection).

STAT: $\{t, L, \rho = L/K_{\text{cap}}, K_{\text{cap}}, C, S, \text{emits}\}$

REG: $\{R_x, R_y, R_z\}$ ENC: $\{B_x, B_y, B_z\}$ SYM: $\{\sigma_x, \sigma_y, \sigma_z\}$

GOV: $\{\lambda, \varepsilon, \gamma, k, \text{overshoot}\}$ INV: $\{\text{budget balance, continuity rms, } |\Omega|\}$

LOG: last 32 events (e.g. $\{t, \text{emit}, \varepsilon, L_{\text{post}}, C_{\text{post}}, S_{\text{post}}\}$)

One-line status: `t=%d L=%.2f rho=%.2f K=%.2f C=%.2f S=%.2f emits=%d |Omega|=%.2f OK`

This card codifies the *disturb* $\rightarrow$ *accumulate* $\rightarrow$ *emit* handshake I conceived: unequal axis budgets bend motion (geometry as encoding), the symmetry gate enforces legal moves, and the governor guarantees finite, inevitable emissions under load.